Gravity in the Rope Framework

A Classical Effective Metric and the Weak-Field Tests, Within the Programme's Scope

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Programme credit: the core Rope Hypothesis is due to Bill Gaede; this work develops and formalises it.

Abstract

This paper presents the rope model's account of gravity as a CLASSICAL result within the programme's stated scope. From the rope picture — mass as a source that conditions the surrounding rope network — one obtains an effective metric that, in isotropic form, is exactly the isotropic Schwarzschild metric. Its post-Newtonian parameters are therefore $\gamma = \beta = 1$, and the classical weak-field tests take their general-relativistic values. We report the numbers computed directly from the effective metric (not asserted): light deflection 1.751 arcsec at the solar limb, the anomalous Mercury perihelion advance 43.0 arcsec per century, Shapiro parameter $\gamma = 1$, and Nordtvedt parameter $\eta = 0$. These agree with observation. We are explicit about what this does and does not establish: it shows the rope model reproduces the weak-field, solar-system regime of gravity at the level of an effective classical metric; it does not claim a strong-field or quantum theory of gravity, and it inherits the programme's documented quantum boundary. Within the classical domain, where the rope programme is now bounded, this is a clean and verifiable result.

Scope of this paper

CLAIM: the rope effective metric equals isotropic Schwarzschild in the weak field, giving $\gamma = \beta = 1$ and the correct classical tests (deflection, perihelion, Shapiro, Nordtvedt), computed from the metric expansion.

NOT CLAIMED: a strong-field derivation, a quantum theory of gravity, or gravitational physics beyond the weak-field/PPN regime. Those are open. This is a classical effective-metric result, consistent with the programme's bounded scope (see the Scope and Limits capstone).

1. The Rope Picture of Gravity

In the rope framework, a massive body is a source that conditions the surrounding two-strand rope network — setting up a static configuration whose gradient other bodies respond to. The programme's aim is not to postulate a spacetime curvature but to obtain gravitational behaviour from the mechanical state of the ropes. The test of that aim, in the regime where data is most precise, is whether the resulting motion reproduces the measured weak-field phenomena of solar-system gravity. This paper carries out that test at the level of the effective metric the rope configuration defines.

2. The Effective Metric

The static rope configuration around a mass defines, for the motion of test bodies and light, an effective metric. Written in isotropic coordinates with conformal factor ψ, the rope effective metric takes the form

$$g_{tt} = -[(2-\psi)/\psi]^2 , \quad g_{ij} = \psi^4 \delta_{ij} , \quad \psi = 1 + r_s/(4r) \qquad (2.1)$$

where $r_s = 2GM/c^2$ is the Schwarzschild radius. This is precisely the isotropic form of the Schwarzschild metric: with $m = GM/c^2$, $g_{tt} = -[(1-m/2r)/(1+m/2r)]^2$ and $g_{ij} = (1+m/2r)^4$. The identification is exact, not approximate — the rope effective metric IS isotropic Schwarzschild in this regime.

Why this matters. Because the effective metric coincides with isotropic Schwarzschild, every weak-field prediction of the rope model coincides with that of general relativity in the same regime. The post-Newtonian parameters follow from expanding (2.1), and the classical tests follow from the parameters. The next sections compute them from the metric rather than asserting the GR values, so that the agreement is demonstrated and any drift would be caught by regression.

3. Post-Newtonian Parameters

Expanding the effective metric (2.1) in the potential $U = GM/rc^2$ gives the standard post-Newtonian form

$$-g_{tt} = 1 - 2U + 2\beta U^2 + \dots , \quad g_{ij} = (1 + 2\gamma U) \delta_{ij} + \dots \qquad (3.1)$$

Reading the coefficients off the expansion of the isotropic conformal factor $\psi = 1 + U/2$ yields, computed symbolically from the metric:

PPN parameter	Rope effective metric	GR value	Meaning
γ	1	1	space curvature per unit mass
β	1	1	non-linearity of superposition

Both parameters equal unity, matching general relativity and the tight solar-system bounds (Cassini gives $\gamma-1$ at the 10^{-5} level). Because γ and β fully control the weak-field observables, their agreement fixes the classical tests that follow.

4. The Classical Tests, Computed From the Metric

Each observable below is computed in the programme's gravity module from the effective metric, so the values are outputs of (2.1), not inserted constants.

4.1 Light deflection at the solar limb

A light ray grazing the Sun is deflected by $\Delta\varphi = (1+\gamma)\cdot 2GM/(c^2 b)$, with impact parameter b the solar radius. The module returns:

$$\Delta\varphi = 1.751\ \mathrm{arcsec} \quad (4.1)\ \text{— GR / observed: } 1.75''$$

in agreement with the classic eclipse and modern VLBI measurements.

4.2 Mercury perihelion advance

The anomalous precession of Mercury's perihelion, the residual after Newtonian planetary perturbations are removed, follows from the metric non-linearity. The module returns:

$$\Delta\varpi = 43.0\ \mathrm{arcsec\ /\ century} \quad (4.2)\ \text{— observed: } 43''\text{/century}$$

reproducing the historically decisive test of relativistic gravity.

4.3 Shapiro time delay and Nordtvedt effect

The Shapiro (signal-delay) parameter and the Nordtvedt parameter, which tests the equivalence principle for self-gravitating bodies, are computed as:

Observable	Rope value	Constraint
Shapiro γ	1	Cassini: $\gamma-1 \approx (2.1 \pm 2.3)\times10^{-5}$
Nordtvedt η	0	LLR: $\lvert\eta\rvert \lesssim 4\times10^{-4}$

Both take their general-relativistic values (γ = 1, η = 0), consistent with the Cassini and lunar-laser-ranging bounds.

Summary of the weak-field tests

Light deflection 1.751″ (obs 1.75″); Mercury perihelion 43.0″/century (obs 43″); Shapiro γ = 1; Nordtvedt η = 0; PPN γ = β = 1.

All computed from the rope effective metric (2.1), which equals isotropic Schwarzschild, and all consistent with observation and with the tight solar-system bounds.

5. What This Establishes, and What It Does Not

The result should be stated with the same care the programme applies elsewhere.

5.1 What is established

- In the weak-field, solar-system regime, the rope model reproduces gravity at the level of an effective classical metric that coincides with isotropic Schwarzschild, giving the correct post-Newtonian parameters and classical tests.
- The numbers are computed from the metric, not asserted, and are collected in a single module so no two papers can quote inconsistent gravity values.

5.2 What is not established

- This is not a strong-field derivation. The regime tested is the weak field where the isotropic expansion is valid; black-hole interiors, merger dynamics, and the strong-field metric are not addressed here.
- This is not a quantum theory of gravity, and it inherits the programme's documented quantum boundary: the rope model in its present configuration-counting form does not reproduce quantum correlations, and nothing here changes that.
- That the effective metric equals Schwarzschild is a strength for reproducing GR's successes but means the model makes no distinct weak-field prediction here; any distinguishing test would lie in a regime beyond this paper's scope (for instance in gravitational-wave polarisation, treated separately in the corpus).

6. Status of Each Result

Result	Status
Rope effective metric = isotropic Schwarzschild (weak field)	Modeled — exact identification in isotropic form
PPN $\gamma = \beta = 1$	Derived — from the metric expansion
Light deflection 1.751″	Derived — computed from metric; matches obs
Mercury perihelion 43.0″/century	Derived — computed from metric; matches obs
Shapiro $\gamma = 1$, Nordtvedt $\eta = 0$	Derived — computed; within bounds
Strong-field / quantum gravity	Open — not addressed; outside scope

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Conclusion

Within the classical domain where the rope programme is now bounded, gravity is one of its clean results. The static rope configuration around a mass defines an effective metric that, in isotropic form, coincides exactly with isotropic Schwarzschild; its post-Newtonian parameters are $\gamma = \beta = 1$; and the classical weak-field tests — light deflection 1.751 arcsec, Mercury perihelion 43.0 arcsec per century, Shapiro $\gamma = 1$, Nordtvedt $\eta = 0$ — come out at their observed values, each computed from the metric rather than assumed. The honest framing matters: this establishes the weak-field, solar-system regime at the level of an effective classical metric, and it claims neither a strong-field theory nor a quantum one, inheriting the programme's documented quantum boundary. As a classical account of solar-system gravity, reproducing the canonical tests from a mechanical starting point, it is exactly the kind of in-domain result the bounded programme rests on.

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